Marmik Shah

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EXPERIENCE

Loylty Rewardz - A BillDesk Group Company *Apr 2025 – Present*
Senior Software Developer *Mumbai, India*

- Architected a channel loyalty product for manufacturers and finance companies, serving agents and channel partners with Hierarchy management, KPI management, leaderboards, tier computations, customer segmentation and an approval module.
- Engineered a real-time, API-driven microservices platform replacing T+1 bulk-file settlement, with 99% of transactions under 800ms; used Spring AOP for cross-cutting concerns and tracing, plus a webhook-based data-relay service.
- Built a low-latency balance service using caching, Bloom filters and database views, handling 300 TPS, letting partners fetch balances and burn points across corporates and Line of business's in a single go.
- Developed a unified vendor integration layer for multiple modules using strategy-based orchestration and Kong gateway, cutting configuration and development effort by 60% when onboarding clients.
- Structured a bucket-based reward matrix framework and optimized enrichment and tenant loading, reducing database calls by over 45% and improving Hikari connection-pool utilization and cutting startup time from 300s to 85s.
- Created AI-assisted admin panels and dev tools using React, so client, product and RM teams manage configurations independently.
- Migrated services to Java 21 and handled SSL/TLS certificate migrations.

Software Developer *Jun 2023 – Mar 2025*

- Engineered a Kafka-driven ingestion platform processing 1M+ records daily, leveraging polling and webhook patterns for asynchronous status tracking and reconciliation to catch missed status updates.
- Designed an enterprise-wide loyalty program letting one program span multiple lines of business, with generic APIs.
- Developed a milestone-based journey rewards product on GraphQL; architected a profile-driven multi-database setup and a strategy driven post-award layer for cashback and other redemption modules.
- Devised a parallel-processing lock for distributed ad hoc transactions and implemented the budget module to control reward spend, with threshold-based actions and real-time budget updates.
- Achieved zero downtime on production services with keep-alive, virtual IPs and master-slave HA; cut Redis memory by over 50% with a purging policy, a cluster setup and tuned RDB snapshots.
- Deployed microservices architecture on a Kubernetes cluster; introduced observability using ELK, Prometheus, and Grafana.

ComED Learning *Apr 2022 – Jul 2022*
Software Development Intern *Remote*

- Built UI/UX components and server-side functionality for the ComED Learning platform using Django and React.
- Connected backend APIs to enable data exchange and dynamic content rendering; resolved application bugs and ran functional tests to make the platform more reliable.

KJ Somaiya College Of Engineering *Feb 2021 – Apr 2021*
Web Development Intern *Mumbai, India*

- Created the website for Symphony, the college cultural fest, collaborating with the cultural and design teams.
- Designed responsive web interfaces featuring interactive components and animations, tuning layouts to fit a range of screen sizes.

PROJECTS

Mumbai Police Portal — Created a full-stack complaint portal with station routing and status tracking (Django, PostgreSQL, Leaflet.js).

Newzer — Developed a personalized news recommendation website that curates and summarizes news articles based on users' interests, preferences, and feedback using various algorithms. (Django, JavaScript, Machine Learning).

Waiterio Application — Implemented a client-server desktop app where waiters place orders and the chef fulfills (Java, Swing, Sockets).

EDUCATION

KJ Somaiya College Of Engineering, *B.Tech in Computer Engineering | CGPI – 8.93* 2019–2023
Maharashtra State Board, *SSC – 89.80% | HSC – 84.62%* 2017–2019

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, TypeScript, Python, SQL, C, C++, Solidity
Frameworks & Libraries: Spring, Spring Boot, Node.js, Django, React, Next.js, Vue.js, SASS, Bootstrap
Databases & Caches: MySQL, PostgreSQL, Greenplum, SQL Server, MongoDB, Supabase, Redis, Neo4j
AI Tools: Claude, GPT, Cursor, Kiro, Copilot
DevOps & Infrastructure: CI/CD, Jenkins, Prometheus, Grafana, New Relic, ELK, Alertmanager, Linux, Bash, Nginx, Crontab
Tools & Platforms: Docker, Kubernetes, Git, Azure, Apache Camel, Airflow, Kong Gateway, IPFS, Parcel
Messaging & APIs: Apache Kafka, RabbitMQ, REST, JPA, GraphQL, JWT, OAuth 2.0
Concepts: Microservices, Distributed Systems, Caching, High Availability, Bloom Filters, Agile, Blockchain, Multi-Tenancy, UI/UX

ACHIEVEMENTS

Awards: Rising Star (2023–24), Extraordinary Diligence (2024–25), Think Out of the Box (2025–26).
Achievements: Won first place in HackerBash (CSI) hackathon as Team GrowMore. Project: Co-Vaccine Web Portal (2021).